Annual data-centre heat reuse for a 500 m³ pool

Simulink R2026a | 8760 hours | synthetic scenario results v02

1. Geometry, thermal capacity and initial heat requirement

Illustrative indoor pool 25 × 12.5 × 1.6 m: V=500 m³ and water surface A=312.5 m². The water is represented by one perfectly mixed thermal state. These dimensions are not proof of competition-pool compliance.

V=L W d = 25 · 12.5 · 1.6 = 500 m³; A=L W = 312.5 m²

m = ρV = 997 · 500 = 498 500 kg

C = ρ V cp = 997 · 500 · 4182 = 2 084 727 000 J/K

E(16→28 °C) = C · 12 / (3.6 · 10^9) = 6.94909 MWh (bez strát / excluding losses)

2. Dynamic energy balance and heat exchange

Algebraic signs matter: convection and radiation can heat the water when the surroundings are warmer. Evaporation depends on activity and cover assumptions. Hall HVAC and physical water circulation are not included.

C · ΔT/Δt = Qpool - Qloss [W]; Δt = 3600 s

Qloss = Qevap + Qconv + Qrad + Qconstruction + Qmakeup + 2000 W

m_evap = 4 · 10^-5 A · max(0, psat(Tw) - RH · psat(Tair)) · activity · (1 - 0.90 cover)

psat(T) = 0.61094 exp(17.625 T / (T + 243.04)) [kPa]

Qevap = m_evap · 2.44 · 10^6 J/kg; Qconv = 3 A (Tw - Tair) (1 - 0.75 cover)

Qrad = 0.95 σ A [(Tw+273.15)^4 - (Tair-2+273.15)^4] (1-0.75 cover)

Qconstruction = 800 (Tw - 20); Qmakeup = m_evap · 4182 · (Tw - 15)

3. Direct heat exchanger, heat pump and dispatch

The direct exchanger depends on temperature approach. A heat pump is used when the source is too cool, with a capped Carnot-based COP scenario model rather than certified manufacturer performance data.

$$Q_{HX} = \min(Q_{demand}, Q_{dc}, 150\ kW, \max(0, 8500\ (T_{source} - T_w - 3)))$$

$$COP = \mathrm{clamp}(0.43 \cdot T_{cond,K} / \max(5, T_{cond,K} - T_{evap,K}), 2.5, 6)$$

$$T_{cond} = \max(34, T_w+8)\ °C;\ T_{evap} = T_{source} - 5\ °C$$

$$Q_{HP} \le \min(Q_{demand}, 150\ kW, Q_{dc} / (1 - 1/COP), 35\ kW \cdot COP)$$

$$P_{HP} = Q_{HP}/COP;\ Q_{source} = Q_{HP} - P_{HP}$$

$$Q_{dc} = Q_{source} + Q_{district} + Q_{storage} + Q_{rejected}\ [W]$$

$$ERF = E_{REUSE} / E_{DC}\ (\text{boundary-specific});\ PUE = E_{DC}/E_{IT}$$

4. Temperature controller, stop signal and time step

The thermostat switches on below 27.35 °C and off above 27.65 °C. An independent model-level heat ceiling prevents the one-hour predicted water temperature exceeding 27.75 °C. STOP routes all available heat to independent rejection instead of the pool, district network or storage.

$$T_{on} = 27.35\ °C;\ T_{off} = 27.65\ °C;\ T_{hard_model} = 27.75\ °C$$

$$Q_{safe} = \max(0, (27.75 - T_w)\ C / \Delta t + Q_{loss})$$

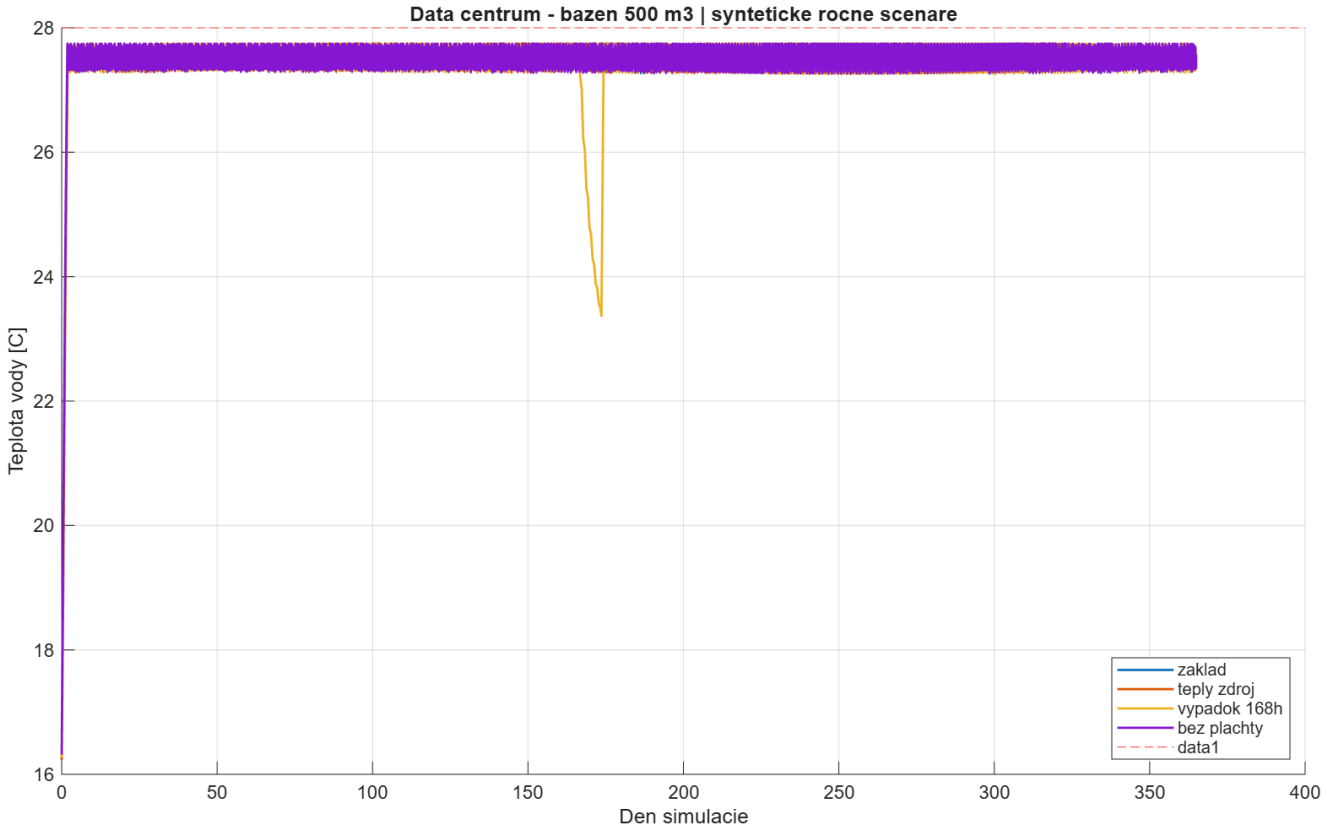
$$Q_{pool} \le \min(150\ kW, Q_{safe});\ T_{new} = T_w + \Delta t\ (Q_{pool} - Q_{loss}) / C$$

$$STOP=1 \Rightarrow Q_{pool}=Q_{district}=Q_{storage}=0;\ Q_{rejected}=Q_{dc}$$

5. Executed 8760-hour scenarios

Four hourly synthetic profiles were run to completion in Simulink. The uncovered scenario removes the nightly pool cover. Numerical figures below are model outputs, not measurements or installation performance promises.

Case	T max [°C]	Pool [MWh]	HP el. [MWh]	DC input [MWh]
Base	27.750	254.39	42.40	211.99
Warmer source	27.750	254.32	28.29	226.03
168 h STOP	27.750	251.74	41.96	209.78
No cover	27.750	331.06	55.18	275.89



6. Uncertainty, verification and design limitations

Source temperature, heat recovery fraction, heat exchanger UA, COP and pool loss parameters are hypothetical. This single-node limiter does not certify sensor fault tolerance or legal compliance. Field design requires measured temperatures and flows, proper hydraulic sizing, equipment COP maps, risk analysis and the applicable consolidated rules.

Verified: data-centre heat allocation, heat pump/heat exchanger energy closure, water storage balance, STOP isolation and maximum modeled water temperature. Reported ERF is only a scenario boundary estimate, not an externally certified KPI.

E_DC,source available = 1629.01 MWh; E_IT = 1894.20 MWh

E_pool = 254.39 MWh = E_DC_to_pool 211.99 + E_HP_el 42.40

E_DC_to_district = 567.69 MWh; E_storage = 53.48 MWh; E_reject = 795.85 MWh

ERF hypothetical / PUE=1.20: 0.343; PUE=1.52: 0.271

Max energy residual (pool) = 4.09e-14 MWh; max water T = 27.750 °C

INPUT PROVENANCE: A = measured, B = published dataset, C = vendor, D = assumed. All operating inputs in these simulations: D.

Primary technical and regulatory references

ASHRAE pool evaporation / natatorium https://handbook.ashrae.org/handbooks/F17/SI/F17_Ch36/f17_ch36_si.aspx
EU Regulation 2024/1364, PUE and ERF https://eur-lex.europa.eu/eli/reg_del/2024/1364/oj
Slov-Lex 308/2012, 2012 reference version https://www.slov-lex.sk/ezbierky/pravne-predpisy/SK/ZZ/2012/308/20121015.html